

Bidirectional Associations Among Family Conflict, Emotion Regulation, and Adolescent Sleep: A Longitudinal Investigation Using Objective Sleep Measurement

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PSYC489: Research Project

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3625 Words

Acknowledgements

I would like to thank my supervisors, Dr. Bart Ellenbroek and Dr. Burt Hatch, for their guidance and support throughout this project. I am particularly grateful to Dr. Hatch for his invaluable help with R coding for the statistical analyses. I also acknowledge the Adolescent Brain Cognitive Development (ABCD) study for making their data available for this research project.

Abstract

This study examined bidirectional associations among family conflict, emotion regulation difficulties, and sleep across early adolescence using longitudinal data from the Adolescent Brain Cognitive Development (ABCD) study. Cross-lagged panel models tested whether family conflict predicted subsequent sleep problems, poor sleep predicted family conflict, and emotion regulation difficulties mediated these relationships. The sample included 884 participants with objective sleep data from Fitbit actigraphy at Years 2 and 4 (ages 11-14), youth-reported family conflict assessments across Years 2-4, and parent-reported emotion dysregulation at Years 3-4. None of the four hypotheses received support. Family conflict did not predict subsequent sleep duration or efficiency, sleep did not predict family conflict, and emotion regulation did not mediate either pathway. Family conflict consistently predicted greater emotion dysregulation one year later, but this association did not extend to sleep outcomes. These null findings contrast sharply with previous research relying on subjective sleep reports, suggesting that associations documented in earlier studies may reflect shared method variance rather than true relationships between objective sleep parameters and psychosocial functioning. The findings challenge theoretical models positioning sleep as a key mechanism linking family stress to adolescent adjustment across annual time lags.

Introduction

Understanding adolescent sleep health represents a critical public health priority, given well-established links between insufficient sleep and adverse outcomes for both physical and mental health (Owens et al., 2014). While research has demonstrated clear associations between family functioning and adolescent sleep patterns, fundamental questions regarding underlying mechanisms remain unanswered. Recent systematic reviews examining longitudinal studies reveal that positive family relational factors, such as parental warmth and cohesion, predict better sleep outcomes, whereas negative factors, especially conflict, predict worse outcomes (Maratia et al., 2023). Importantly, children's own perceptions of family conflict appear particularly critical, as their subjective appraisals more strongly predict developmental outcomes than parent reports (Grych & Fincham, 1990).

Despite this growing evidence base, existing research suffers from significant methodological limitations. Most critically, the field has relied almost exclusively on subjective sleep reports and cross-sectional designs that cannot distinguish temporal ordering or bidirectional effects (Maratia et al., 2023). The present study addresses these limitations by examining bidirectional associations among family conflict, emotion regulation difficulties, and objectively measured sleep across early adolescence using longitudinal data from the Adolescent Brain Cognitive Development (ABCD) study.

Family Conflict and Adolescent Sleep

A large body of evidence has linked family conflict with poorer adolescent sleep across multiple parameters. Cross-sectional studies consistently document that adolescents from high-conflict families report more difficulty falling asleep and shorter sleep durations compared to those from lower-conflict families (Maratia et al., 2023). Longitudinal studies have also begun to clarify the temporal ordering of such effects, with meta-analyses suggesting that family conflict predicts subsequent deterioration in sleep quality over time

(Maratia et al., 2023). These effects appear particularly pronounced during early adolescence, a developmental period characterised by significant changes in both family relationships and sleep patterns, including delayed circadian phases and increased sleep needs (Ge et al., 2025). Multiple mechanisms may explain how family conflict disrupts adolescent sleep. According to emotional security theory, family conflict threatens children's sense of security within the family system, leading to heightened vigilance and physiological arousal that interferes with sleep onset (Davies & Cummings, 1994). Additionally, family conflict may activate stress response systems, resulting in elevated evening cortisol that delays sleep onset and reduces sleep efficiency (Armstrong-Carter et al., 2021). These physiological disruptions may be compounded by the emotional and cognitive disruptions that follow exposure to family conflict, creating a cascade of processes that undermine healthy sleep.

Family Conflict and Emotion Regulation

Parallel lines of research have also linked family conflict with adolescent emotion regulation difficulties. Theoretical frameworks propose that family dynamics fundamentally shape children's developing capacity for emotional self-regulation, the ability to monitor and evaluate emotion reactions in contrast to accomplishing their goals (Cheng et al., 2024). Children exposed to frequent family conflict often exhibit greater difficulties across multiple regulatory dimensions, including limited access to effective regulation strategies and a tendency to experience emotions as overwhelming and uncontrollable (Cheng et al., 2024). Longitudinal evidence supports this temporal priority of family conflict in predicting subsequent emotion regulation difficulties. Studies done during early adolescence demonstrate that exposure to inter-parental conflict predicts declines in emotion regulation capacity over time (Hansen, 2017). Additionally, the family emotional climate has been shown to predict later emotion dysregulation even when controlling for earlier regulatory difficulties, suggesting true developmental effects rather than mere stability of individual

differences (Speidel et al., 2020). These effects likely emerge through multiple pathways. Children in high-conflict families have reduced opportunities to observe and internalise adaptive emotional responses, as parents' model poor emotional control during conflicts (Cheng et al., 2024). Family conflict may also create emotionally invalidating environments where children's emotional experiences are dismissed or criticised, undermining the development of emotional awareness, a key component of effective emotion regulation (Seddon et al., 2020).

Emotion Regulation and Sleep

Research has also established clear connections between emotion regulation difficulties and sleep problems in adolescents, with emerging research suggesting bidirectional effects. Adolescents who struggle to regulate emotions effectively exhibit poorer sleep quality and shorter sleep durations (Feghali et al., 2023). Emotion regulation difficulties are hypothesised to operate as transdiagnostic mechanisms linking interpersonal stress with health outcomes. Specifically, adolescents who experience emotions as overwhelming may experience prolonged arousal at bedtime, interfering with the natural transition to sleep (Sesso et al., 2024). Rumination, the persistent cognitive focus on distressing experiences, has been specifically linked to sleep onset difficulties and reduced sleep efficiency in adolescent samples (Wang et al., 2025). Importantly, this relationship may be bidirectional rather than unidirectional. Experimental sleep restriction studies demonstrate that sleep loss impairs prefrontal cortex functioning involved in the cognitive control of emotion, leading to heightened emotional reactivity and difficulty implementing adaptive regulation strategies (Baum et al., 2013). Adolescents may be particularly vulnerable to these effects given the ongoing maturation of the prefrontal-limbic circuits characteristic of this developmental period (Ladouceur, 2012). This potential bidirectional relationship raises the

possibility for cascading effects whereby family conflict disrupts both emotion regulation and sleep, which then mutually reinforce each other over time.

Mediating Role of Emotion Regulation

Despite the well-established associations between family conflict and sleep, family conflict and emotion regulation, and emotion regulation and sleep, no research has directly investigated whether emotion regulation difficulties mediate the relationship between family conflict and adolescent sleep during early adolescence. Although theoretical models suggest that family stress disrupts emotional self-regulation, which subsequently affects sleep patterns, this proposed mediation pathway lacks direct empirical validation. Testing this mediation hypothesis is critical for several reasons. First, identifying emotion regulation as a mediator would clarify mechanisms and provide support for theoretical models proposing that family environments influence adolescent health outcomes through effects on developing self-regulatory capacities. Second, demonstrating mediation would inform intervention design by suggesting that programs targeting either family conflict or emotion regulation skills might yield benefits for adolescent sleep health. Clarifying these mechanisms is essential for both advancing developmental theory and designing evidence-based interventions.

Methodological Limitations of Existing Research

While prior research has established important associations, existing studies suffer from methodological limitations that constrain causal inferences and our understanding of these relationships. Most critically, the literature has relied predominantly on subjective sleep reports rather than objective measurements (Sadeh, 2011). This reliance is problematic, as research using ABCD data has demonstrated that parent-reported adolescent sleep durations significantly overestimate objective actigraphy data (Turan et al., 2024), suggesting that subjective reports may capture perceptions of sleep rather than actual sleep behaviour. When

the same informant then reports on multiple constructs, associations may be inflated by shared method variance, correlations arising from common measurement methods rather than true relationships between the underlying constructs. Objective measures such as actigraphy offer substantial advantages over subjective reports, including elimination of recall bias, continuous monitoring in naturalistic environments, and more reliable estimates of specific sleep parameters such as sleep duration and efficiency (Lucas-Thompson et al., 2020). These benefits substantially improve both the reliability and ecological validity of sleep assessment in adolescent samples.

A second critical limitation is the predominance of cross-sectional designs in this literature. Cross-sectional approaches cannot investigate bidirectionality, leaving unresolved the possibility that poor sleep not only results from family conflict but also exacerbates conflict by increasing irritability in both adolescents and parents. While longitudinal studies demonstrate that family conflict predicts later sleep problems, whether poor sleep contributes to increased family dysfunction through emotional or behavioural pathways has received minimal investigation (Maratia et al., 2023). This gap is particularly significant because reciprocal feedback loops are especially relevant during adolescence, a developmental stage characterised by heightened sensitivity to interpersonal stress and ongoing neural maturation (Blakemore & Mills, 2013). Family systems theory emphasises that adolescent development operates bidirectionally, suggesting that disrupted sleep could fuel escalating cycles of parent-child conflict (Cox & Paley, 2003). Understanding such reciprocal relationships is critical for designing targeted interventions, particularly given that poor adolescent sleep has been linked to immune impairment, metabolic dysregulation, and increased risk for chronic health conditions (Knutson et al., 2007). Interventions that break these cycles could therefore yield benefits across multiple domains of adolescent health and functioning.

The ABCD Study

The ABCD study represents one of the largest longitudinal investigations of adolescent development, with 11,878 participants initially recruited at ages 9-10 and designed to reflect the demographic diversity of the United States (Garavan et al., 2018). The study provides comprehensive sleep assessment through Fitbit actigraphy data collected from approximately 4,000 participants over 21-day periods, alongside validated assessments of family functioning and emotion regulation (Garavan et al., 2018). This combination of objective sleep measurement and longitudinal design spanning ages 9 to 14 provides a unique opportunity to address the methodological limitations that have constrained previous research. The timing of ABCD assessments is particularly advantageous, as this period encompasses substantial changes in both family relationships and sleep patterns. Early adolescence is characterised by normative shifts in circadian rhythms, hormonal changes, and increased autonomy from parents, all factors that may increase susceptibility to the effects of family stresses on sleep (Zhang et al., 2024). By examining these relationships during this critical developmental window using objective measurement, the ABCD study enables more rigorous tests of theoretical models linking family environments to adolescent sleep health.

The Present Study

The present study utilises ABCD data to examine bidirectional associations among family conflict, emotion regulation difficulties, and sleep across early adolescence. Using objective actigraphy-derived sleep measures and cross-lagged panel models, we test four primary hypotheses: First, does family conflict predict subsequent changes in sleep? Specifically, we predict that higher youth-reported family conflict (FES Conflict subscale) at Year 3 (ages 12-13) will predict lower objectively measured sleep duration and lower sleep efficiency at Year 4 (ages 13-14). Second, does poor sleep predict subsequent increases in family conflict? Specifically, we predict that lower objectively measured sleep duration and

lower sleep efficiency at Year 2 (ages 11-12) will predict higher youth-reported family conflict at Year 3. Third, does emotion regulation difficulties mediate the path from family conflict to sleep? Specifically, we predict that higher youth-reported family conflict at Year 2 will predict greater parent-reported emotion dysregulation (DERS Catastrophising subscale) at Year 3, which will in turn predict lower objectively measured sleep duration and lower sleep efficiency at Year 4. Fourth, does emotion regulation difficulties mediate the path from sleep to family conflict? Specifically, we predict that lower objectively measured sleep duration and lower sleep efficiency at Year 2 will predict greater parent-reported emotion dysregulation at Year 3 which will in turn predict higher youth-reported family conflict at Year 4.

We examine these relationships separately for sleep duration and sleep efficiency to determine whether effects differ across parameters that capture different aspects of sleep health, with duration reflecting total sleep quantity and efficiency reflecting sleep quality and continuity. Critically, these parameters can vary independently. Adolescents may maintain adequate sleep duration yet experience fragmented, inefficient sleep characterised by frequent awakenings, or conversely, they may achieve high sleep efficiency with insufficient sleep duration (Kim et al., 2022). Research demonstrates that sleep duration and efficiency show only modest correlations and predict different developmental outcomes (Thompson et al., 2024). Therefore, by investigating both parameters separately, we can determine whether family conflict and emotion regulation show specific associations with sleep quantity versus sleep quality, providing a more comprehensive and nuanced understanding of how family dynamics relate to different aspects of adolescent sleep health. By testing these hypotheses using rigorous longitudinal methods and objective sleep assessment, this study addresses critical gaps in the literature and provides novel insights into the mechanisms linking family environments to adolescent sleep health.

Method

Participants

Data were drawn from the Adolescent Brain Cognitive Development study, a longitudinal study of adolescent development in the United States. The ABCD study recruited 11,878 children aged 9-10 years at baseline from 21 sites across the country. Participants were recruited primarily through probability sampling of public and private schools with defined catchment areas for each study site, with a smaller proportion recruited through community-based methods including summer camps and word-of-mouth referrals (Garavan et al., 2018). This recruitment strategy was designed to approximate the sociodemographic diversity of the United States population. The present analyses utilised data from Year 2 (ages 11-12), Year 3 (ages 12-13), and Year 4 (ages 13-14) assessments. The sample was restricted to participants who were either the oldest child or only child in their family to avoid non-independence of observations from siblings. Because of this 2176 twins and 934 younger siblings were removed from the sample. Additionally, participants were required to have at least 7 days of valid sleep data at both Year 2 and Year 4, as well as complete data on family conflict across Years 2, 3, and 4, and emotion dysregulation at Years 3 and 4. After applying these criteria, the final sample included 884 participants. Sample demographic characteristics are presented in Table 1. The sample did not significantly differ from excluded participants on sex, $\chi^2(2) = 3.07, p = .22$, or household income, $\chi^2(9) = 10.07, p = .35$, but did differ on race/ethnicity, $\chi^2(4) = 75.01, p < .001$, with the sample including significantly more Asian (4.2% vs. 2.0%) and Hispanic (25.7% vs. 19.9%) participants and less Black participants (6.9% vs. 15.7%).

Table 1*Sample Demographic Characteristics*

Characteristic	Category	%
Sex		
	Female	48.4
	Intersex Male	0.1
	Male	51.5
Race/Ethnicity		
	Asian	4.2
	Black	6.9
	Hispanic	25.7
	Other	11.1
	White	52.1
Household Income		
	\$200,000 and greater	12.3
	\$100,000 through \$199,999	32.5
	\$75,000 through \$99,999	15.3
	\$50,000 through \$74,999	13.6
	\$35,000 through \$49,999	8.4
	\$25,000 through \$34,999	5.8
	\$16,000 through \$24,999	4.9
	\$12,000 through \$15,999	1.7
	\$5,000 through \$11,999	13.6
	Less than \$5,000	2.5

Note. The sample did not significantly differ from excluded participants on sex, $\chi^2(2) = 3.07, p = .22$, or household income, $\chi^2(9) = 10.07, p = .35$, but did differ on race/ethnicity, $\chi^2(4) = 75.01, p < .001$, with the sample including significantly more Asian and Hispanic participants and less Black participants.

Materials

Sleep patterns were assessed using the Fitbit Charge HR 2 or 3 wearable device worn continuously for up to 21 days at each assessment wave. Participants were instructed to wear

the device 24 hours per day, removing it only for water activities. Nightly sleep data was considered valid if sleep duration was above 3 hours and below 15 hours. Participants were required to have at least 7 days of valid sleep data at both Year 2 and Year 4 for inclusion in analyses. When a day was excluded due to invalid data, the next available valid day replaced it in the sequence. Sleep duration was calculated as the total number of minutes of sleep per night. Sleep efficiency was calculated as the percentage of time in bed spent asleep (total sleep time / time in bed x 100). In our primary analyses, we used nightly sleep data of the first seven valid nights of recordings as indicators for latent variable estimates at Year 2 and Year 4. This approach has been used previously (Hatch et al., 2019). Unlike simply calculating an arithmetic average, using a latent variable means we get an estimate of sleep across multiple nights after partialling out night-to-night variation. While night-to-night variation is of interest for some research questions, the current research questions relate to individual differences in sleep over multiple nights.

Family conflict was assessed using the Conflict subscale of the Family Environment Scale, which consists of 9 items assessing the amount of openly expressed anger and aggression among family members (FES; Moos & Moos, 1994). Youth rated each item on a true/false scale. Sample items include “Family members often criticise each other” and “We fight a lot in our family”.

Emotion dysregulation was assessed using the Catastrophise subscale of the parent-report version of the Difficulties in Emotion Regulation Scale (DERS-P; Bunford et al., 2020). The DERS-P was adapted from the original adolescent self-report DERS (Gratz & Roemer, 2004) to assess parents’ perceptions of their child’s emotion regulation difficulties. The Catastrophise subscale consists of 11 items that assess the child’s tendency to lose control and feel overwhelmed by negative emotions. Items are rated on a 5-point Likert scale ranging from 1 (*almost never*) to 5 (*almost always*), with higher scores indicating greater

emotion dysregulation. Sample items include “My child experiences his/her emotions as overwhelming and out of control” and “When my child is upset, he/she loses control over his/her behaviours”.

Procedure

The ABCD study procedures were approved by a centralised Institutional Review Board at the University of California, San Diego, as well as by local IRBs at each of the 21 data collection sites. Parents or legal guardians provided written informed consent, and youth provided written assent prior to participation. All procedures complied with the Declaration of Helsinki guidelines for ethical research with human participants. Data collection occurred annually beginning at baseline (ages 9-10) through young adulthood. The present study utilised data from Year 2 (November 2018 to November 2020, ages 11-12), Year 3 (July 2019 to January 2021, ages 12-13), and Year 4 (November 2020 to November 2022, ages 13-14) assessments. At each annual assessment wave, families completed questionnaires either during in-person visits to their local ABCD study site or via a secure online portal. Youth independently completed the Family Environment Scale, while parents independently completed the DERS-P regarding their child’s emotion regulation. All questionnaires were administered in English or Spanish based on the respondent’s preference.

At Years 2 and 4, a subset of participants was invited to participate in objective sleep monitoring using Fitbit Charge HR 2 or 3 wearable devices. Approximately 5,900 participants at Year 2 and 4,000 participants at Year 4 were provided with Fitbit devices. Research staff provided detailed instructions on device use and troubleshooting during the initial visit. Participants were instructed to wear the device continuously for up to 21 days on their non-dominant wrist, removing it only for water activities such as bathing or swimming. The Fitbit app was downloaded to either the parent’s or child’s cell phone to enable automatic data syncing when the device was in close proximity to the phone. Research staff

monitored data syncing remotely, and families were contacted if a child's Fitbit data had not been uploaded for four consecutive days. Following the 21-day monitoring period, participants returned devices by mail to the study site. Participants and their families received monetary compensation for their time and effort at each assessment wave, typically provided in the form of cash or gift cards, with the amount varying by site and assessment wave but designed to be equal to the burden of participation. It should be noted that data collection procedures during the Year 3 and Year 4 assessments were affected by the COVID-19 pandemic. During the period from March 2020 onward, many in-person assessments were converted to remote or virtual formats to comply with public health guidelines. However, Fitbit devices continued to be distributed to participants' homes via mail, and the sleep monitoring protocol remained largely unchanged.

Design

This study employed a longitudinal correlational design examining bidirectional associations among family conflict, emotion regulation difficulties, and sleep across three annual assessment waves (Years 2, 3, and 4). Cross-lagged panel models were used to test temporal precedence and bidirectional effects. Two parallel models were tested: one with sleep duration as the sleep outcome and one with sleep efficiency as the sleep outcome.

Data Analysis

Prior to testing the primary hypotheses, measurement invariance was evaluated for all constructs to ensure that the same latent constructs were being measured consistently across time. For sleep duration and sleep efficiency, configural, metric (equal factor loadings), scalar (equal intercepts), and strict (equal residual variances) invariance models were tested across Years 2 and 4. For emotion dysregulation and family conflict, which used ordinal indicators, threshold invariance (equal thresholds), metric invariance, and residual invariance were tested. Measurement invariance was evaluated using chi-square difference tests and changes

in approximate fit indices. Following recommendations by Cheung and Rensvold (2002) and Chen (2007), changes in CFI (ΔCFI) $\leq .010$ and changes in RMSEA (ΔRMSEA) $\leq .015$ were considered evidence of invariance. Models were estimated using robust maximum likelihood estimation (MLR) for continuous indicators and robust weighted least squares (WLSMV) for ordinal indicators in *lavaan* version 0.6-20 (Rosseel, 2012) in R version 4.5.2 (R Core Team, 2025).

Cross-lagged panel models (CLPMs) were estimated to examine bidirectional associations among sleep, family conflict, and emotion dysregulation across Years 2, 3, and 4. Two separate models were tested: one with sleep duration and one with sleep efficiency. Each CLPM included autoregressive paths representing the stability of each construct from one wave to the next, cross-lagged paths representing temporal predictions between constructs, and within-time covariances among all constructs at each wave. Latent variables were created for each construct at each time point. Sleep duration and sleep efficiency were each measured by 7 daily observations. Family conflict was measured by 9 FES Conflict subscale items. Emotion dysregulation was measured by 11 DERS Catastrophise subscale items. To ensure measurement invariance across time, factor loadings were constrained to equality across waves for each construct based on the measurement invariance results. Residual covariances were estimated for the same indicators across adjacent waves. For measures with ordinal indicators (FES and DERS), the average loading constraint method was used for identification, where the average of all factor loadings was constrained to equal the number of indicators (Wu & Estabrook, 2016).

Following recommendations for analysing ABCD baseline data (Dick et al., 2021), we did not include additional demographic covariates (e.g., race/ethnicity, socioeconomic status) beyond the study design features already accounted for in our models. Dick et al. (2021) note that for regression models of ABCD baseline data, mixed-effects models that

account for nesting of participants within families and sites represent the preferred analytical approach, and there is currently no empirical evidence that including additional covariates improves model accuracy or precision. Additionally, because variables such as socioeconomic status and other demographics may serve as proxies for both potential confounders and mediators or colliders simultaneously, their inclusion without clear theoretical justification can obscure rather than clarify the relationships of interest (Dick et al., 2021).

Models were estimated using robust weighted least squares with mean and variance adjustment (WLSMV) estimation to account for the ordinal nature of the FES and DERS items. Missing data were handled using pairwise deletion. Across the variables included in the models, Little's test of data missing completely at random was not significant, indicating that data were not missing from each variable systematically (Little, 1988). Model fit was evaluated using multiple indices: chi-square test statistic (χ^2), comparative fit index (CFI), Tucker-Lewis index (TLI), root mean square error of approximation (RMSEA) with 90% confidence intervals, and standardised root mean square residual (SRMR). Following guidelines by Hu and Bentler (1999), good fit was indicated by $CFI \geq .95$, $TLI \geq .95$, $RMSEA \leq .06$, and $SRMR \leq .09$. Adequate fit was indicated by $CFI \geq .90$, $TLI \geq .90$, $RMSEA \leq .08$, and $SRMR \leq .10$. Standardised path coefficients (β) are reported for all structural paths. Statistical significance was evaluated at $p < .05$. All analyses were conducted in R version 4.5.2 (R Core Team, 2025) using the *lavaan* package version 0.6-20 (Rosseel, 2012) and *semTools* package version 0.5-7 (Jorgensen et al., 2025) for measurement invariance testing.

Results

Measurement Invariance Testing

Measurement invariance was assessed for all constructs across time (see Appendix A, Tables A1-A4). For sleep duration across Years 2 and 4 (see Table A1), the configural model demonstrated acceptable fit, with metric invariance supported showing minimal change in indices ($\Delta\text{CFI} = -.002$, $\Delta\text{RMSEA} = -.001$). Scalar invariance was also supported ($\Delta\text{CFI} = .003$, $\Delta\text{RMSEA} = -.002$), while strict invariance was not supported ($\Delta\text{CFI} = -.114$, $\Delta\text{RMSEA} = .014$). Therefore, scalar invariance was retained, indicating that factor loadings and intercepts were equivalent across time, allowing for meaningful comparisons of latent means.

For sleep efficiency across Years 2 and 4 (see Table A2), the configural model demonstrated excellent fit, with both metric invariance ($\Delta\text{CFI} = -.003$, $\Delta\text{RMSEA} = -.002$) and scalar invariance ($\Delta\text{CFI} = .003$, $\Delta\text{RMSEA} = .003$) supported, though strict invariance was not supported ($\Delta\text{CFI} = -.013$, $\Delta\text{RMSEA} = .008$). Therefore, scalar invariance was retained, indicating that factor loadings and intercepts were equivalent across time, allowing for meaningful comparisons of latent means.

For emotion dysregulation across Years 3 and 4 (see Table A3), the configural model demonstrated excellent fit, with threshold invariance ($\Delta\text{CFI} = .000$, $\Delta\text{RMSEA} = -.004$), metric invariance ($\Delta\text{CFI} = .000$, $\Delta\text{RMSEA} = -.001$), and residual invariance ($\Delta\text{CFI} = .000$, $\Delta\text{RMSEA} = .000$) all supported. Therefore, residual invariance was retained, indicating that thresholds, factor loadings, and residual variances were equivalent across time, allowing for meaningful comparisons of latent means and supporting the strongest level of measurement invariance.

For family conflict across Years 2, 3, and 4 (see Table A4), the configural model demonstrated good fit, with threshold invariance supported ($\Delta\text{CFI} = -.005$, $\Delta\text{RMSEA} = .003$). Therefore, threshold invariance was retained, indicating that thresholds and factor loadings were equivalent across time, allowing for meaningful comparisons of latent means.

Sleep Duration Cross-Lagged Panel Model

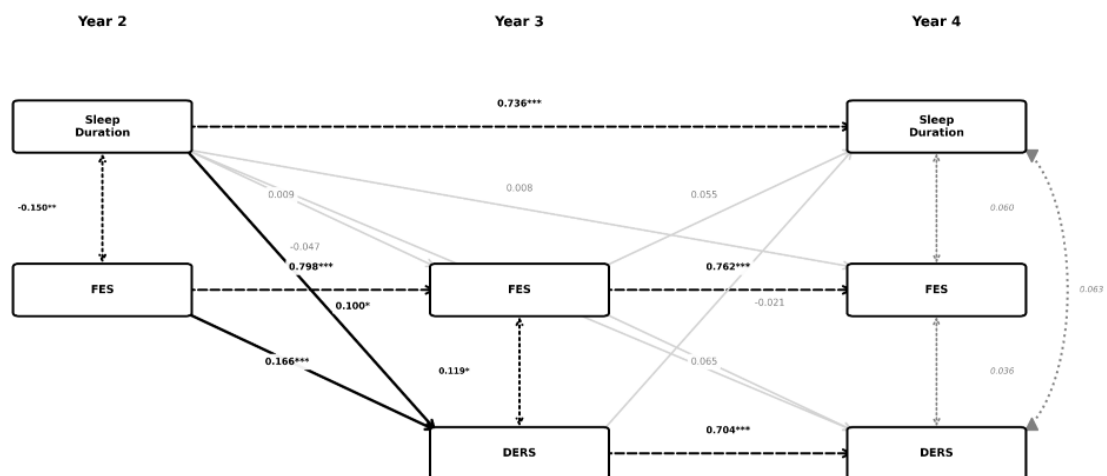
Figure 1 presents the cross-lagged panel model for sleep duration, family conflict, and emotion dysregulation with standardised path coefficients (see Table 2 for complete standardised path coefficients and significance tests). The sleep duration CLPM demonstrated good fit to the data. Although the chi-square test was significant, which is common with large sample sizes, the approximate fit indices indicated adequate model fit: CFI = .953, TLI = .949, RMSEA = .029 (90% CI [.028, .031]), SRMR = .069. These indices suggest that the model adequately represented the observed data. All constructs demonstrated significant stability across time, with sleep duration showing high stability from Year 2 to Year 4 ($\beta = .736, p < .001$), family conflict showing very high stability from Year 2 to Year 3 ($\beta = .798, p < .001$), and from Year 3 to Year 4 ($\beta = .762, p < .001$), and emotion dysregulation showing high stability from Year 3 to Year 4 ($\beta = .704, p < .001$). These findings indicate that individual differences in each construct were largely maintained across time.

Several significant cross-lagged associations emerged. Unexpectedly, greater sleep duration at Year 2 significantly predicted increased emotion regulation difficulties at Year 3 ($\beta = .100, p = .030$). Additionally, higher family conflict at Year 2 predicted greater emotion dysregulation at Year 3 ($\beta = .166, p < .001$). However, sleep duration at Year 2 did not significantly predict family conflict at Year 3 ($\beta = .009, p = .832$). At Year 4, neither sleep duration at Year 2 nor family conflict at Year 3 significantly predicted emotion dysregulation ($\beta = -.047, p = .219$, and $\beta = .065, p = .070$, respectively). Sleep duration at Year 2 also did not significantly predict family conflict at Year 4 ($\beta = .008, p = .869$). Similarly, family conflict at Year 3 did not predict sleep duration at Year 4 ($\beta = .055, p = .321$), and emotion dysregulation at Year 3 did not predict sleep duration at Year 4 ($\beta = -.021, p = .672$). Emotion dysregulation at Year 3 also did not significantly predict family conflict at Year 4 ($\beta = .057, p = .124$). Regarding within-time associations, at Year 2, sleep duration was negatively

correlated with family conflict ($r = -.150, p = .003$), indicating that shorter sleep was associated with higher family conflict. At Year 3, family conflict and emotion dysregulation were positively correlated ($r = .119, p = .037$). At Year 4, the within-time covariances among sleep duration, family conflict, and emotion dysregulation were all not significant (all $p > .05$).

Figure 1

Cross-Lagged Panel Model Examining Bidirectional Associations Between Sleep Duration, Family Conflict, and Emotion Dysregulation Across Years 2, 3, and 4



Note. Standardised path coefficients are shown. Dashed lines represent autoregressive paths, solid black lines represent significant cross-lagged paths, and light gray lines represent non-significant cross-lagged paths. Dotted lines indicate covariances. * $p < .05$. ** $p < .01$. *** $p < .001$.

Table 2

Standardised Path Coefficients for Cross-Lagged Sleep Duration Model

Autoregressive Paths	β	p
Sleep Y2 → Sleep Y4	.736	< .001
FES Y2 → FES Y3	.798	< .001
FES Y3 → FES Y4	.762	< .001
DERS Y3 → DERS Y4	.704	< .001
Cross-Lagged Paths		

Sleep Y2 → DERS Y3	.100	.030
Sleep Y2 → FES Y3	.009	.832
FES Y2 → DERS Y3	.166	< .001
Sleep Y2 → DERS Y4	-.047	.219
Sleep Y2 → FES Y4	.008	.869
FES Y3 → DERS Y4	.065	.070
FES Y3 → Sleep Y4	.055	.321
DERS Y3 → Sleep Y4	-.021	.672
DERS Y3 → FES Y4	.057	.124
Covariances		
Sleep Y2 ↔ FES Y2	-.150	.003
FES Y3 ↔ DERS Y3	.119	.037
Sleep Y4 ↔ FES Y4	.060	.513
Sleep Y4 ↔ DERS Y4	.036	.511
FES Y4 ↔ DERS Y4	.063	.397

Note. FES = Family Environment Scale (Conflict subscale); DERS = Difficulties in Emotion Regulation Scale (Catastrophising subscale). All coefficients are standardised. Significant paths ($p < .05$) are bolded in the table.

Sleep Efficiency Cross-Lagged Panel Model

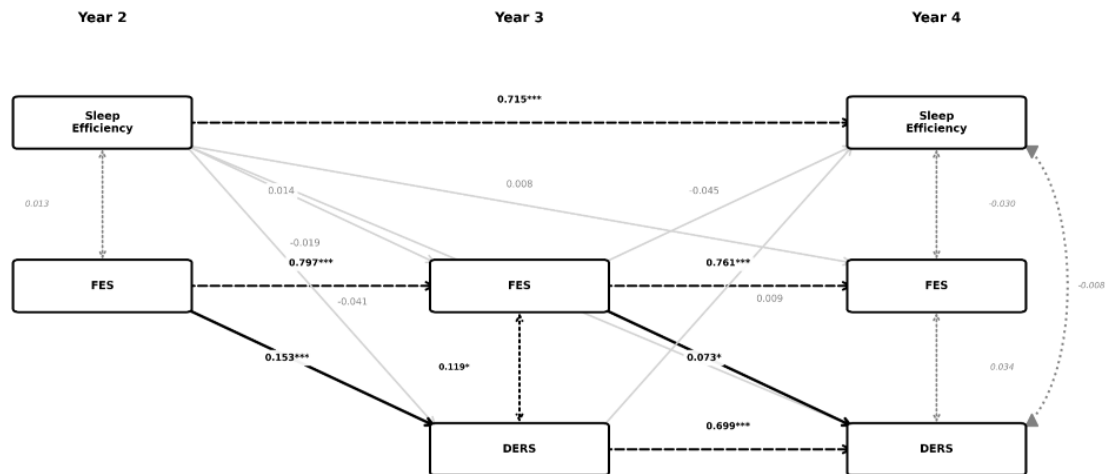
Figure 2 presents the cross-lagged panel model for sleep efficiency, family conflict, and emotion dysregulation with standardised path coefficients (see Table 3 for complete standardised path coefficients and significance tests). The sleep efficiency CLPM demonstrated good fit to the data: CFI = .952, TLI = .949, RMSEA = .030 (90% CI [.028, .031]), SRMR = .069. These fit indices suggest that the model adequately captured the relationships among sleep efficiency, family conflict, and emotion dysregulation. All constructs demonstrated significant stability across time, with sleep efficiency showing high stability from Year 2 to Year 4 ($\beta = .715, p < .001$), family conflict showing very high stability from Year 2 to Year 3 ($\beta = .797, p < .001$), and from Year 3 to Year 4 ($\beta = .761, p < .001$), and emotion dysregulation showing high stability from Year 3 to Year 4 ($\beta = .699, p$

<.001). These autoregressive coefficients were comparable to those observed in the sleep duration model.

The pattern of cross-lagged effects differed slightly from the sleep duration model. At Year 3, sleep efficiency at Year 2 did not significantly predict emotion dysregulation ($\beta = -.041, p = .324$). However, higher family conflict at Year 2 significantly predicted greater emotion dysregulation at Year 3 ($\beta = .153, p < .001$). Sleep efficiency at Year 2 did not predict family conflict at Year 3 ($\beta = .014, p = .705$). At Year 4, sleep efficiency at Year 2 did not predict emotion dysregulation ($\beta = -.019, p = .517$). However, family conflict at Year 3 significantly predicted emotion dysregulation at Year 4 ($\beta = .073, p = .042$), suggesting that higher family conflict was associated with increases in emotion dysregulation across time. Sleep efficiency at Year 2 did not predict family conflict at Year 4 ($\beta = .008, p = .838$). Family conflict at Year 3 did not predict sleep efficiency at Year 4 ($\beta = -.045, p = .283$), and emotion dysregulation at Year 3 did not predict sleep efficiency at Year 4 ($\beta = .009, p = .828$). Emotion dysregulation at Year 3 also did not significantly predict family conflict at Year 4 ($\beta = .058, p = .112$). Regarding within-time associations, at Year 2, the correlation between sleep efficiency and family conflict was not significant ($r = .013, p = .767$). At Year 3, family conflict and emotion dysregulation were positively correlated ($r = .119, p = .040$). At Year 4, none of the within-time covariances among sleep efficiency, family conflict, and emotion dysregulation were significant (all $p > .05$).

Figure 2

Cross-Lagged Panel Model Examining Bidirectional Associations Between Sleep Efficiency, Family Conflict, and Emotion Dysregulation Across Years 2, 3, and 4



Note. Standardised path coefficients are shown. Dashed lines represent autoregressive paths, solid black lines represent significant cross-lagged paths, and light gray lines represent non-significant cross-lagged paths. Dotted lines indicate covariances. * $p < .05$. ** $p < .01$. *** $p < .001$.

Table 3

Standardised Path Coefficients for Cross-Lagged Sleep Efficiency Model

Autoregressive Paths	β	p
Sleep Y2 → Sleep Y4	.715	< .001
FES Y2 → FES Y3	.797	< .001
FES Y3 → FES Y4	.761	< .001
DERS Y3 → DERS Y4	.699	< .001
Cross-Lagged Paths		
Sleep Y2 → DERS Y3	-.041	.324
Sleep Y2 → FES Y3	.014	.705
FES Y2 → DERS Y3	.153	< .001
Sleep Y2 → DERS Y4	-.019	.517
Sleep Y2 → FES Y4	.008	.838
FES Y3 → DERS Y4	.073	.042
FES Y3 → Sleep Y4	-.045	.283

DERS Y3 → Sleep Y4	.009	.828
DERS Y3 → FES Y4	.058	.112
Covariances		
Sleep Y2 ↔ FES Y2	.013	.767
FES Y3 ↔ DERS Y3	.119	.040
Sleep Y4 ↔ FES Y4	-.030	.677
Sleep Y4 ↔ DERS Y4	.034	.541
FES Y4 ↔ DERS Y4	-.008	.886

Note. FES = Family Environment Scale (Conflict subscale); DERS = Difficulties in Emotion Regulation Scale (Catastrophising subscale). All coefficients are standardised. Significant paths ($p < .05$) are bolded in the table.

Discussion

Summary of Key Findings

None of our four hypotheses received empirical support. Hypothesis 1, that family conflict would predict subsequent sleep problems, was not supported as family conflict did not predict either sleep duration or sleep efficiency at any time point. Hypothesis 2, that poor sleep would predict subsequent family conflict, was similarly unsupported. Hypotheses 3 and 4, which proposed that emotion dysregulation would mediate the relationships between family conflict and sleep (in both directions), were also not supported, as the necessary sequential associations involving sleep were absent. While family conflict at Year 2 consistently predicted greater emotion dysregulation at Year 3 across both models, this association was not part of our primary hypotheses, all of which centrally involved sleep as either a predictor or outcome.

Interpreting the Null Findings

The one significant sleep-related finding, that sleep duration at Year 2 predicted emotion dysregulation at Year 3, requires careful interpretation, as it almost certainly represents a statistical suppression effect rather than a genuine predictive relationship. A suppression effect occurs when the relationship between two variables becomes significant

only when a third variable is included in the model, even though the bivariate association is nonsignificant (Watson et al., 2013). The significant path coefficient only emerged when family conflict was simultaneously included in the cross-lagged panel model. This pattern indicates that the apparent effect of sleep duration on emotion dysregulation is entirely contingent upon the presence of family conflict in the model. In other words, sleep duration shows no relationship with subsequent emotion regulation difficulties when examined in isolation, the relationship materialises only through the specific variance partitioning that occurs in the multivariate model. This substantially undermines any interpretation of this finding as evidence for a causal effect of sleep on emotion regulation development.

The comprehensive null findings for sleep relationships represent a stark contrast to previous literature suggesting associations between family functioning and adolescent sleep. A critical factor in understanding this discrepancy concerns the use of objective versus subjective sleep measures. Previous research linking sleep to family functioning and emotional outcomes has relied predominantly on subjective sleep reports from either parents or adolescents. However, research using the ABCD dataset has demonstrated that parent-reported adolescent sleep durations significantly overestimate objective sleep data derived from actigraphy, with subjective reports showing substantial bias (Turan et al., 2024). This discrepancy suggests that subjective sleep reports may capture something fundamentally different from actual sleep parameters, potentially reflecting parents' or adolescents' general perceptions of functioning or response biases rather than actual sleep behaviour.

Another particularly important consideration concerns the source of measurement across studies in this literature. When parents report on both their child's sleep problems and emotion regulation difficulties, correlations may be artificially elevated due to the shared method variance, associations that arise from common measurement method rather than true covariation in the underlying constructs (Zimmer-Gembeck et al., 2021). Parents

experiencing high levels of family stress may be particularly likely to perceive multiple domains of their child's functioning negatively, creating spurious associations between constructs (Neece et al., 2012). Similarly, adolescents experiencing negative affect may report both more sleep problems and more family conflict, inflating associations through shared report bias rather than true relationships. In the present study, emotion dysregulation was assessed via parent report, family conflict via youth report, and sleep objectively via actigraphy. This multi-method approach eliminates shared method variance and provides a more conservative, and arguably more valid, test of the relationships between sleep and psychosocial functioning. The comprehensive null findings obtained with objective sleep measurement suggest that previous findings based on subjective reports may have been substantially inflated by shared method variance. This interpretation receives strong support from the meta-analysis of Maratia and colleagues (2023) where the three studies using objective actigraphy failed to find significant associations between family factors and sleep, whereas studies using subjective measures consistently reported such associations. Our findings replicate and extend this pattern with a substantially larger sample and more rigorous longitudinal design, strengthening the conclusion that objective and subjective sleep measures may capture fundamentally different constructs. This does not mean that subjective sleep perceptions are not important, how individuals perceive and interpret their sleep may be more proximally related to emotional and interpersonal functioning than objective sleep parameters. However, it does suggest that much of the previous literature may have documented associations between perceptions of problems across domains rather than true relationships between actual sleep behaviour and family or emotional functioning.

The most consistent finding across both models was that higher family conflict at Year 2 predicted greater emotion dysregulation at Year 3. This association aligns with developmental theories proposing that family environments shape children's emerging

emotion regulation capacities through modelling of emotional responses and the creation of emotionally validating environments (Yang et al., 2023). However, the effect sizes were relatively modest, particularly when considered alongside the very high stability coefficients for family conflict and emotion dysregulation. This pattern suggests that while family conflict does predict subsequent changes in emotion regulation, individual differences in both constructs are predominantly maintained across this developmental period. The modest cross-lagged effects reflect the limited variance available for prediction after accounting for substantial stability, a common challenge in developmental research examining constructs with high rank-order stability (Cheng et al., 2024). Importantly, this finding does not salvage support for our hypotheses, which all centrally involve sleep. Rather, it indicates that family conflict and emotion regulation are meaningfully connected during early adolescence, while sleep, at least as objectively measured, appears disconnected from these psychosocial processes over the timescales examined.

Limitations

The inability to detect hypothesised effects, particularly the mediation pathways, raises important questions about the adequacy of our statistical approach. While we employed three waves of data, the minimum required for testing mediation in a cross-lagged framework, substantial methodological literature has demonstrated that three-wave mediation models face significant limitations (Reichardt, 2011). Critically, three-wave designs cannot adequately separate within-person change from between-person differences, a distinction essential for drawing causal inferences about mediation (Cain et al., 2017). With only three waves, cross-lagged panel models conflate stable between-person differences (e.g., adolescents who consistently experience high family conflict also consistently show poor emotion regulation) with within-person change (e.g., increases in family conflict lead to subsequent increases in emotion regulation difficulties). This confounding may obscure true

mediational processes or create spurious ones. More sophisticated approaches, such as random-intercept cross-lagged panel models, can separate these components but require four or more waves of data to be properly identified and to provide stable estimates (Falkenström et al., 2022). With four waves, we could model stable trait-like differences between individuals separately from within-person fluctuations over time, providing a more rigorous test of whether changes in family conflict led to changes in emotion regulation, which in turn lead to changes in sleep. Additional waves would also enable more flexible modelling of temporal dynamics, such as testing whether effects occur with shorter or longer lags, and whether there are delayed or cumulative effects that unfold over multiple measurement occasions. The inability to model these complexities with our current three-wave design represents an important limitation. However, given the comprehensive absence of sleep-related effects at even the bivariate level, it seems unlikely that more sophisticated modelling would reveal the hypothesised relationships. The problem appears to be not just one of statistical modelling, but of genuine absence of associations between objective sleep parameters and psychosocial functioning over annual time lags.

Our sample differed significantly from the excluded ABCD participants in racial/ethnic composition, with fewer Black participants and more Asian and Hispanic participants. This differential attrition likely reflects the requirements for valid actigraphy data and complete assessments across multiple waves, which may be more challenging for families experiencing socioeconomic disadvantage or instability, factors that disproportionately affect Black families in the United States due to systemic inequities (Mindlis et al., 2020). This differential attrition has important implications for generalisability. The relationships among family conflict, emotion regulation, and sleep may differ across racial/ethnic groups due to cultural differences in family dynamics (Karcher et al., 2022). Additionally, families experiencing greater adversity may show stronger effects of

family stress on child functioning, as chronic stress can deplete regulatory resources and exacerbate the impact of additional stressors (Nweze et al., 2022). Our reduced representation of Black participants and potentially lower-resourced families may have resulted in an underestimation of true associations in the broader population. However, it is important to note that even with these sample restrictions, our analysis included nearly 900 participants with objective sleep data, a substantially larger sample than has been available in previous research using actigraphy. Most previous studies examining family factors and adolescent sleep with objective measurement have relied on samples of fewer than 250 participants (Evans et al., 2021). Our sample size provided adequate statistical power to detect small-to-moderate effects, making it unlikely that the null findings simply reflect insufficient power. The combination of a large sample size, objective measurement, and longitudinal design represents a significant advance in methodological rigour, strengthening confidence in the null findings.

Implications and Future Directions

These comprehensive null findings carry important theoretical implications. They challenge existing models that position sleep as a key mechanism linking family stress to adolescent adjustment, at least when sleep is measured objectively and examined over annual time lags. Either these theoretical models require substantial revision, or the relationships operate over different timescales than examined here, or they reflect primarily subjective perceptions rather than objective sleep behaviour. From a methodological perspective, these findings highlight the critical importance of measurement methods in developmental research. The stark contrast between our null findings with objective sleep measurement and previous positive findings with subjective reports suggest that a substantial portion of the existing literature may have documented shared method variance rather than true substantive

relationships. This underscores the necessity of multi-method assessment and the dangers of relying exclusively on single-reporter data.

Future research should address several key limitations of the present study. First, studies should employ four or more waves of data to enable random-intercept cross-lagged panel models that can separate between-person from within-person effects. Second, research should examine multiple timescales, using both intensive longitudinal designs to capture short-term dynamics and longer follow-up periods to examine developmental cascades. Third, studies should systematically compare objective and subjective sleep measures to determine whether they predict different outcomes and whether subjective perceptions mediate relationships between objective sleep parameters and functioning. Such future research will be essential for further clarifying whether and how sleep relates to psychosocial development during adolescence.

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Appendix A

Measurement Invariance Testing Results

Table A1

Measurement Invariance Model for Sleep Duration Years 2 and 4

Model	χ^2	df	CFI	TLI	RMS EA	[90% CI]	SRM R	$\Delta\chi^2$	Δdf	ΔCFI	ΔRM SEA
Confi gural	126.9 7	69	.919	.894	0.31	[.023, .039]	.032				
Metric 3	134.0 3	75	.918	.901	.030	[.022, .038]	.034	7.06	6	-.002	-.001
Scala r	138.0 9	81	.921	.911	.028	 [.020, .036]	.035	4.06	6	.003	-.002
Strict 5	226.8 5	88	.807	.801	.042	[.035, .049]	.051	88.76	7	-.114	.014

Note. All χ^2 difference tests used robust maximum likelihood estimation (MLR). Bolded model represents the most parsimonious model demonstrating adequate invariance. CFI = comparative fit index; TLI = Tucker-Lewis index; RMSEA = root mean square error of approximation; CI = confidence interval; SRMR = standardised root mean square residual. Invariance criteria: $\Delta CFI \leq .010$ and $\Delta RMSEA \leq .015$ (Chen, 2007; Cheung & Rensvold, 2002). Sleep duration demonstrated scalar invariance, indicating equivalence of factor loadings and intercepts across time.

Table A2

Measurement Invariance Model for Sleep Efficiency Years 2 and 4

Model	χ^2	df	CFI	TLI	RMS EA	[90% CI]	SRM R	$\Delta\chi^2$	Δdf	ΔCFI	ΔRM SEA
Confi gural	85.67	69	.991	.988	.018	[.007, .026]	.023				
Metric	96.86	75	.988	.986	.018	[.009, .027]	.029	11.19	6	-.003	.000
Scala r	98.26	81	.991	.989	.016	 [.007, .025]	.029	1.40	6	.003	-.002
Strict 9	129.5 9	88	.977	.976	.023	[.015, .031]	.035	31.33	7	-.013	.007

Note. All χ^2 difference tests used robust maximum likelihood estimation (MLR). Bolded model represents the most parsimonious model demonstrating adequate invariance. CFI = comparative fit index; TLI = Tucker-Lewis index; RMSEA = root mean square error of approximation; CI = confidence interval; SRMR = standardised root mean square residual. Invariance criteria: $\Delta CFI \leq .010$ and $\Delta RMSEA \leq .015$ (Chen, 2007; Cheung & Rensvold,

2002). Sleep efficiency demonstrated scalar invariance, indicating equivalence of factor loadings and intercepts across time.

Table A3

Measurement Invariance Model for Emotion Dysregulation Years 3 and 4

Model	χ^2	df	CFI	TLI	RMS EA	[90% CI]	SRM R	$\Delta\chi^2$	Δ df	Δ CFI	Δ RM SEA
Confi gural 3	933.8 3	197	.993	.992	.067	[.062, .071]	.059				
Thres hold 4	943.7 4	219	.993	.993	.063	[.059, .068]	.059	9.91	22	.000	-.004
Metric 4	948.3 4	229	.993	.993	0.62	[.058, .067]	.059	4.60	10	.000	-.001
Resid ual 4	993.3 4	240	.993	.993	.062	 [.058, .067]	.060	45.00	11	.000	.000

Note. All models estimated using robust weighted least squares with mean and variance adjustment (WLSMV) for ordinal indicators. Bolded model represents the most parsimonious model demonstrating adequate invariance. CFI = comparative fit index; TLI = Tucker-Lewis index; RMSEA = root mean square error of approximation; CI = confidence interval; SRMR = standardised root mean square residual. Invariance criteria: Δ CFI $\leq$.010 and Δ RMSEA $\leq$.015 (Chen, 2007; Cheung & Rensvold, 2002). Emotion dysregulation demonstrated residual invariance (strong invariance), indicating equivalence of thresholds, factor loadings, and residual variances.

Table A4

Measurement Invariance Model for Family Conflict Years 2, 3, and 4

Model	χ^2	df	CFI	TLI	RMS EA	[90% CI]	SRM R	$\Delta\chi^2$	Δ df	Δ CFI	Δ RM SEA
Confi gural 67	1274. 67	845	.982	.979	.024	[.021, .027]	.077				
Thres hold 23	1422. 23	873	.977	.974	.027	 [.024, .029]	.079	147.5 6	28	-.005	.003
Metric 09	1473. 09	891	.975	.973	.027	[.025, .030]	.080	50.86	18	-.002	.000
Resid ual 32	1658. 32	909	.969	.968	.030	[.028, .033]	.085	185.2 3	18	-.006	..003

Note. All models estimated using robust weighted least squares with mean and variance adjustment (WLSMV) for ordinal indicators. Bolded model represents the most parsimonious model demonstrating adequate invariance. CFI = comparative fit index; TLI = Tucker-Lewis index; RMSEA = root mean square error of

approximation; CI = confidence interval; SRMR = standardised root mean square residual. Invariance criteria: $\Delta\text{CFI} \leq .010$ and $\Delta\text{RMSEA} \leq .015$ (Chen, 2007; Cheung & Rensvold, 2002). Family conflict demonstrated threshold invariance, indicating equivalence of thresholds and factor loadings across time.